

WEEK – 3

RELATIONAL SCHEMA DESIGN REPORT

Project Title

ShopNest – E-Commerce Order
Management Database System

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1. Introduction

The **ShopNest – E-Commerce Order Management Database System** uses a relational database to store and manage business information efficiently. After designing the Entity Relationship (ER) Diagram in Week 3, the next step is converting the conceptual model into a Relational Schema.

A relational schema represents each entity as a table and defines the relationships between tables using **Primary Keys (PK)** and **Foreign Keys (FK)**. It provides the logical structure required for implementing the database using SQL while maintaining data integrity and reducing redundancy.

2. Relational Schema Design

CUSTOMER

```
CUSTOMER
(
  CUSTOMER_ID (PK),
  NAME,
  EMAIL,
  MOBILE_NUMBER,
  ADDRESS,
  REGISTRATION_DATE
)
```

CATEGORY

```
CATEGORY
(
  CATEGORY_ID (PK),
  CATEGORY_NAME,
  DESCRIPTION
)
```

SUPPLIER

```
SUPPLIER
(
  SUPPLIER_ID (PK),
  SUPPLIER_NAME,
  CONTACT_INFORMATION
)
```

PRODUCT

```
PRODUCT
(
  PRODUCT_ID (PK),
  PRODUCT_NAME,
  PRICE,
  STOCK_QUANTITY,
  CATEGORY_ID (FK),
  SUPPLIER_ID (FK)
)
```

ORDER

```
ORDER
(
  ORDER_ID (PK),
  CUSTOMER_ID (FK),
  ORDER_DATE,
  TOTAL_AMOUNT,
  ORDER_STATUS
)
```

ORDER_DETAILS

```
ORDER_DETAILS
(
  ORDER_DETAIL_ID (PK),
  ORDER_ID (FK),
  PRODUCT_ID (FK),
  QUANTITY,
  UNIT_PRICE
)
```

PAYMENT

```
PAYMENT
(
  PAYMENT_ID (PK),
  ORDER_ID (FK),
  PAYMENT_METHOD,
  PAYMENT_DATE,
  PAYMENT_STATUS
)
```

SHIPMENT

```
SHIPMENT
(
  SHIPMENT_ID (PK),
  ORDER_ID (FK),
  DELIVERY_ADDRESS,
  SHIPMENT_DATE,
  DELIVERY_STATUS
)
```

REVIEW

```
REVIEW
(
  REVIEW_ID (PK),
  CUSTOMER_ID (FK),
  PRODUCT_ID (FK),
  RATING,
  COMMENTS
)
```

3. Table Structure

Table Name	Primary Key	Foreign Key(s)
CUSTOMER	CUSTOMER_ID	—
CATEGORY	CATEGORY_ID	—
SUPPLIER	SUPPLIER_ID	—
PRODUCT	PRODUCT_ID	CATEGORY_ID, SUPPLIER_ID
ORDER	ORDER_ID	CUSTOMER_ID
ORDER_DETAILS	ORDER_DETAIL_ID	ORDER_ID, PRODUCT_ID
PAYMENT	PAYMENT_ID	ORDER_ID
SHIPMENT	SHIPMENT_ID	ORDER_ID
REVIEW	REVIEW_ID	CUSTOMER_ID, PRODUCT_ID

4. Relationship Mapping

Parent Table	Child Table	Foreign Key	Cardinality
CUSTOMER	ORDER	CUSTOMER_ID	One-to-Many (1:M)
CATEGORY	PRODUCT	CATEGORY_ID	One-to-Many (1:M)
SUPPLIER	PRODUCT	SUPPLIER_ID	One-to-Many (1:M)
ORDER	ORDER_DETAILS	ORDER_ID	One-to-Many (1:M)
PRODUCT	ORDER_DETAILS	PRODUCT_ID	One-to-Many (1:M)
ORDER	PAYMENT	ORDER_ID	One-to-One (1:1)
ORDER	SHIPMENT	ORDER_ID	One-to-One (1:1)
CUSTOMER	REVIEW	CUSTOMER_ID	One-to-Many (1:M)
PRODUCT	REVIEW	PRODUCT_ID	One-to-Many (1:M)

5. Relational Schema Summary

The relational schema converts all nine entities identified in the ER Diagram into relational tables. Each table contains a unique **Primary Key (PK)**, while related tables are connected using **Foreign Keys (FK)** to maintain referential integrity.

The schema supports customer management, product categorization, supplier management, order processing, payment tracking, shipment management, and customer reviews. The relationships between tables ensure accurate data storage, reduce redundancy, and support efficient SQL implementation.

6. Conclusion

The **Relational Schema Design** provides the logical structure required for implementing the **ShopNest – E-Commerce Order Management Database System** in a relational database management system. All entities, attributes, primary keys, and foreign keys have been properly mapped from the ER Diagram into relational tables.

This relational schema serves as the foundation for the next phase of the project, including **database creation, SQL table implementation, normalization, insertion of sample data, query execution, and report generation.**